WASTEWISE-ML · TRACEABILITY MATRIX

WasteWise-ml traceability matrix

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DOCUMENT CONTROL

Document ID	WML-RTM-001	Version	0.2
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/03b-traceability.md @ 9031e25		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

WasteWise-ml traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 23 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-013	PR-001	Material class accuracy	WML-CAL-001, WML-BLD-001, WML-STM-001	Test on the field evaluation set, TRL 4 (on hold)	At risk
UR-002	PR-002	Plastic grade accuracy from a photo	WML-CAL-001, WML-STM-001, WML-OPT-001	Test on the field evaluation set, TRL 4	Not met
UR-003	PR-003	Hazard flag	WML-CAL-001, WML-STM-001	Test on the field evaluation set, TRL 4; design review of the app	At risk
UR-004	PR-004	Abstain when unsure	WML-CAL-001, WML-STM-001	Coverage and accuracy curve on the field set, TRL 4 (66 % on paper, WML-CAL-001 section 8)	At risk
UR-005, UR-008	PR-005	Offline, on-device speed	WML-CAL-001, WML-STM-001, WML-OPT-001	Analysis (WML-CAL-001 section 3: 201 ms from a preview-size frame, up to 331 ms if a full 12 MP frame is decoded), TRL 3; timing on the reference phone, TRL 4	At risk
UR-008	PR-006	Small download	WML-CAL-001, WML-STM-001	Analysis (WML-CAL-001 section 2), TRL 3; file sizes of the exports, TRL 4	Met on paper
UR-005	PR-007	Quick to use	WML-CAL-001, WML-STM-001	Time budget (WML-CAL-001 section 4), TRL 3; timed trials with users, TRL 4	To verify at TRL 4
UR-006	PR-008	Usable without reading	WML-CAL-001, WML-BLD-001	Co-design sessions and usability trials, TRL 4	To verify at TRL 4
UR-010	PR-009	Local grades	WML-CAL-001	Design review, TRL 3	Met by design
UR-009	PR-010	Battery life	WML-CAL-001, WML-BLD-001, WML-STM-001	Energy estimate (WML-CAL-001 section 5), TRL 3; measurement, TRL 4	Met on paper
UR-013	PR-011	Field evaluation set	WML-CAL-001, WML-STM-001	Dataset manifest, TRL 4	Not met
UR-008, UR-016	PR-012	Reference phone	WML-CAL-001	Analysis (BOM line 1), TRL 3; device test, TRL 4	Met on paper
UR-012	PR-013	Open and licensed	WML-CAL-001, WML-STM-001	License review, TRL 3 and 4	At risk
UR-011	PR-014	Consent and privacy	WML-CAL-001, WML-BLD-001, WML-OPT-001	Data protocol review with the partner, TRL 4	To verify at TRL 4
UR-007	PR-015	Scan trigger at hand	WML-CAL-001, WML-BLD-001, WML-STM-001	Inspection of the model (WML-CAL-001 section 12: 45 mm), TRL 3; first checks, TRL 4	Met by design
	PR-016				

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-006, UR-007		Result legible from where users stand	WML-CAL-001, WML-BLD-001	Analysis (WML-CAL-001 section 12: 43 arcmin at the 5th percentile, but at 83 degrees off the screen normal), TRL 3; trial with a short user, TRL 4	To verify at TRL 4
UR-015	PR-017	Mat sealed all over	WML-CAL-001, WML-BLD-001	Inspection (WML-BLD-001 section 3.7), TRL 3 and 4	Met by design
UR-003	PR-018	Loose cells cannot short	WML-CAL-001, WML-STM-001	Inspection; site briefing, TRL 4	Met by design
UR-001	PR-019	Picture on the mat	WML-CAL-001, WML-BLD-001, WML-STM-001, WML-OPT-001	Analysis (WML-CAL-001 section 6: 439 mm at 26 mm plus 3.3 mm a side of stack; fits lenses of 25.76 mm and longer), TRL 3; first check "Picture on the mat", TRL 4	At risk
UR-015	PR-020	Phone temperature	WML-CAL-001, WML-STM-001	Analysis (WML-CAL-001 section 13: 40.7 C), TRL 3; site temperature log, TRL 4	Met on paper
UR-014, UR-016	PR-021	Built locally with hand tools	WML-CAL-001	Inspection of the build plan (WML-BLD-001), TRL 3	Met by design
UR-017	PR-022	Phone lifts out	WML-CAL-001, WML-BLD-001	First check "Phone in and out" (WML-BLD-001), TRL 4	To verify at TRL 4
UR-001	PR-023	Stand rigid	WML-CAL-001, WML-BLD-001	Analysis (WML-CAL-001 section 11: 0.30 mm), TRL 3; first check "Stand rigid", TRL 4	Met on paper

Gaps

None.